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### Personal Information

- **First name:** Isabel María
- **Last name:** Moreno Cuadrado
- **Date of birth:** 1 June 2001
- **Place of origin:** Granada, Spain

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### Education

- 2025–2029 **PhD in Advanced Mathematics**,  
(expected) *Institute of Mathematical Sciences (ICMAT), CSIC*, Madrid, Spain.
  - Research area: **Quantum Information Theory**.
  - Position: **CSIC Severo Ochoa Programme Researcher**.
- 2024–2025 **MSc in Advanced Mathematics**,  
*Complutense University of Madrid*, Madrid, Spain.
  - Research area: **Quantum Information Theory**.
- 2019–2024 **BSc in Mathematics**,  
*University of Granada*, Granada, Spain.
  - Particular interest in Mathematical Analysis and Applied Mathematics.
  - *Honours* in the Bachelor's Thesis.
- 2019–2024 **BSc in Computer Science**,  
*University of Granada*, Granada, Spain.
  - Specialisation in Artificial Intelligence.
  - *Honours* in the Bachelor's Thesis.
- 2017–2019 **High School Diploma**, *Regina Mundi School*, Granada, Spain.

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### Research Experience

- **Introduction to Research Programme (PI3).**  
June 2025 – September 2025, IMUS, Seville, Spain.  
Supervisor: Prof. Julián Eduardo Haddad
  - Fourier transform, distributions, and convex geometry: from the Busemann–Petty problem to modern functional analysis.
- **IAE INTRO ICU Research Contract.**  
September 2024 – July 2025, CSIC, Spain.  
*Quantum Information Mathematics: foundations and applications*, ICMAT.  
Supervisors: Prof. David Pérez and Carlos Palazuelos
  - Bohnenblust–Hille inequality and applications in quantum learning theory.
- **Severo Ochoa Programme (INTRO-SO).**  
June 2024 – July 2024, ICMAT, Madrid, Spain.  
Supervisor: Prof. Manuel Garzón Martínez
  - Distribution theory and applications to PDEs.
- **ADIA Lab International Summer School.**  
July 2024, University of Granada in collaboration with Abu Dhabi University.
  - Summer school/conference on trustworthy, secure, and governable AI.
  - Second prize for the project “Learning from Small Data”.
- **Undergraduate Collaboration Grant.**  
October 2023 – July 2024, Department of Mathematical Analysis, University of Granada.  
Supervisor: Prof. Francisco Javier Meri de la Maza
  - Research on distribution theory and its application to neural networks.

- **Summer Research Programme.**

July 2023 – September 2023, Donostia International Physics Center (DIPC), San Sebastián, Spain.

Supervisor: Prof. Thomas Frederiksen

- Application of mathematical and physical models to develop an app to explore the electronic structure of nanographene.

## Awards and Scholarships

- **Blas Cabrera Summer School Scholarship**

August 2025, Menéndez Pelayo International University.

Scholarship programme supporting research initiation, teaching, and scientific innovation for 80 top-achieving university students.

- **Blas Cabrera Summer School Scholarship**

August 2024, Menéndez Pelayo International University.

- **Banco Santander Scholarship.**

September 2023 – June 2024, University of Granada.

- **UGR Skills Acceleration Programme.**

September 2023 – May 2024, Incubadora de Talento, University of Granada.

Selective programme offering training and professional development opportunities to 20 top-ranked students across disciplines.

## Projects

- **Master's Thesis (with Honours).**

September 2024 – June 2025, Complutense University of Madrid, Madrid, Spain.

- Study of the Bohnenblust–Hille inequality and its applications in learning theory and quantum information theory.  
Supervisors: [Prof. David Pérez](#) and [Carlos Palazuelos](#).

- **Bachelor's Thesis (with Honours).**

September 2023 – June 2024, University of Granada, Granada, Spain.

- Study of Fourier analysis in  $\mathbb{R}^n$ . Supervisor: [Prof. Francisco Javier Meri de la Maza](#).
- Analysis of the Fast Fourier Transform (FFT) to accelerate the training of Convolutional Neural Networks.  
Supervisor: [Prof. Jesús Giráldez Crú](#).

- **Parking Space Detection Project in Granada.**

September 2023 – January 2024, Computer Vision course, University of Granada, Granada, Spain.

- Final project for the Computer Vision course.
- GitHub repository: [granada-parking-segmentation](#).

- **Fraud Detection in Financial Transactions Project.**

March 2024 – May 2024, Machine Learning course, University of Granada, Granada, Spain.

- Final project for the Machine Learning course.

## Courses and Academic Activities

- **Attendance at the 7th BYMAT Conference: Bringing Young Mathematicians Together.**

- Talk titled “The Bohnenblust–Hille inequality and its applications in learning theory”.

- **Invited to the Young Researchers Workshop.**

22–24 September 2025, UCM, Madrid.

- Presentation titled “Bohnenblust–Hille inequality and applications in learning theory”.

- **IAE Mathematics Summer School.**

June–July 2024 and June–July 2025, ICMAT, Madrid, Spain.

- Summer school organised annually by ICMAT for undergraduate and master's students interested in mathematical research and its applications.

- **National Meeting of Mathematics Students (ENEM).**

21–25 July 2025, Granada.

- Poster presentation titled “Bohnenblust–Hille inequalities and applications in learning theory”.
- Talk titled “What can pure mathematics tell us about how we learn?”.
- Informational session on academic careers.

- **Attendance at the conference “Quantum Technologies for Young Researchers”.**

8–11 July 2025, Madrid.

- Poster presentation titled “Bohnenblust–Hille inequalities and applications in quantum information theory”.

- **Meeting of the Mathematical Analysis and Applications Network.**  
Cullera, 3–7 March 2025.
  - Conference attendance.
  - Research project and presentation: “*Almost disjoint families*”.
- **Attendance at the Biennial RSME Congress 2025.**  
Bilbao, 13–17 January 2025.
  - Attendance at Angela Capel (University of Cambridge) plenary lecture titled “*Many-Body Quantum Systems*”.
  - Attendance at the Quantum Information Theory parallel session.
  - Poster presentation titled “*Fast Fourier Transform*”.
  - Talk titled “*Fourier analysis in computer science*”.
- **Instructor for an Artificial Intelligence Workshop.**  
29 November, University of Granada, Granada, Spain.
  - Delivered a workshop on AI concepts and applications for undergraduate and master’s students.
- **Attendance at the 6th BYMAT Conference: Bringing Young Mathematicians Together.**
  - Attendance at the Quantum parallel session.
  - Poster presentation titled “Behind the algorithm: the historical paper that shaped modern signal processing”.
  - Talk titled “Computer science through the frequency spectrum”.
- **Attendance at the National Meeting of Mathematics Students (ENEM).**
  - July 2022, University of Valencia.
  - July 2023, University of Badajoz.

## Community Involvement

- Class representative for five consecutive years.
- Member of the academic committee of the Mathematics degree programme.
- Speaker at science outreach events and school talks.
- Active participation in academic and social volunteering initiatives.

## Languages

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|---------|---------------------|
| Spanish | <b>Native</b>       |
| English | <b>Advanced</b>     |
| Chinese | <b>Intermediate</b> |
| French  | <b>Beginner</b>     |